Sharp Convergence and Sample Complexity of Policy Mirror Descent for Average-Reward MDPs

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Discounted vs. average reward

Discounted

$$\underbrace{\rho_\gamma = \lim_{H\to\infty} \sum_{k=0}^{H-1} \gamma^k c_k}_{\text{more weight on early costs}}$$

- initial conditions matter
- effective horizon $H_\gamma := 1/(1-\gamma)$

Average reward

$$\underbrace{\rho = \lim_{H\to\infty} \frac{1}{H} \sum_{k=0}^{H-1} c_k}_{\text{equal weight on all costs}}$$

- stationary behavior matters
- no horizon: only mixing (t_{mix})

Policy mirror descent

$$\pi_s^{t+1} \in \arg\min_{p \in \mathcal{P}(\mathcal{A})} \{\eta_t \langle \tilde{Q}_s^t, p \rangle + D(p, \pi_s^t)\}$$

Tabular critic	$\tilde{Q}^t(s,a) = \hat{Q}^t(s,a)$
LFA critic	$\tilde{Q}^t(s,a) = \phi_{s,a}^\top w_t, \;\; w_t \in \mathbb{R}^d$

How fast, and with how few samples, can PMD provably minimize the average-reward objective?

Master recursion

Lemma 3.7. If the mismatch $\chi_t := \|d^{\pi^\star}/d^{\pi^t}\|_\infty \le \bar{\chi}$ for all t and the step sizes grow as $\eta_t/\eta_{t-1} \ge \bar{\chi}/(\bar{\chi}-1)$, then for any critic with errors at most $\bar{A}$, $\bar{B}$ and all $t \ge 0$,

$$\big(\Delta_t - \underbrace{(\bar{\chi}\bar{A} + \bar{B})}_{\text{error floor}}\big)_+ \le \underbrace{\Big(1 - \frac{1}{\bar{\chi}}\Big)^t}_{\text{linear rate}} \cdot \underbrace{\Psi_0}_{\text{initialization}}$$

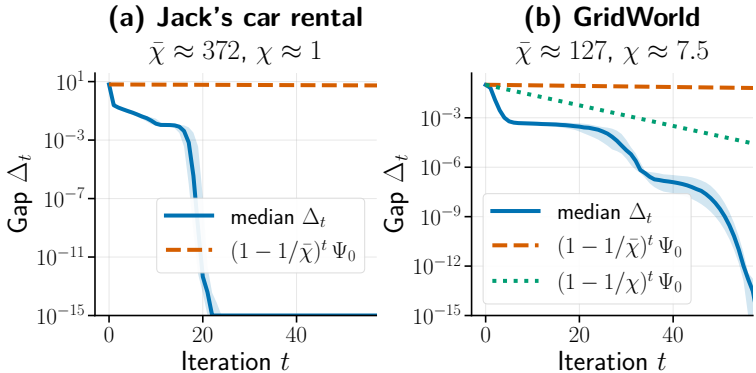
If $\eta_t/\eta_{t-1} \to \infty$, then for every $\zeta > 0$ and the tail mismatch $\chi := \limsup_t \chi_t \le \bar{\chi}$,

$$\limsup_{t\to\infty} \big(\Delta_t - ((\chi+\zeta)\bar{A} + \bar{B})\big)_+^{1/t} \le 1 - \frac{1}{\chi+\zeta}$$

- **Actor and critic separate.** The actor sets the rate, the critic only the error floor.
- **No tuning needed.** Doubling the step size every iteration works without knowing $\bar{\chi}$.

Exact critic

With an exact critic, $\bar{A} = \bar{B} = 0$: the error floor vanishes and $\limsup_t \Delta_t^{1/t} \leq 1 - 1/\chi$.



Tabular critic

By Lemma 3.7, $\Delta_T \le \varepsilon$ follows from:

- **Actor.** $T = \tilde{O}(\bar{\chi})$ iterations, with one critic call each.
- **Critic.** Errors $\bar{A}, \bar{B} = O(\varepsilon/\bar{\chi})$, so that the error floor satisfies $\bar{\chi}\bar{A} + \bar{B} \le \varepsilon$.

Roll out once per state, not per action.

$$Q^\pi(s,a) = c(s,a) - \rho(\pi) + \underbrace{\sum_{s'} P(s' \mid s,a)}_{\text{one-step samples for every } (s,a)} \cdot \underbrace{h^\pi(s')}_{\text{rollouts once per state}}$$

Prior best PG: $|\mathcal{S}||\mathcal{A}|\, t_{\text{mix}}^3 \longrightarrow |\mathcal{S}|\, t_{\text{mix}}^3 + |\mathcal{S}||\mathcal{A}|\, t_{\text{mix}}^2$

LFA critic

Average cheap pointwise estimates, not expensive critic estimates.

$$w_{t,k} = w_{t,k-1} - \alpha\big(\phi_{s_k,a_k}^\top w_{t,k-1} - \underbrace{\hat{Q}^t(s_k,a_k)}_{\text{noisy label from one rollout}}\big)\phi_{s_k,a_k}$$

$$w_t = \underbrace{\frac{1}{\lceil K/2 \rceil}\sum_{k>K/2} w_{t,k}}_{\text{averaging over SGD steps}}$$

Prior best PG: $|\mathcal{S}||\mathcal{A}|\, t_{\text{mix}}^5 \longrightarrow d\, t_{\text{mix}}^3$

Sample complexity

Samples to reach $\Delta_T \leq \varepsilon$ (log factors omitted, $\times 1/\varepsilon^2$):

	Tabular generative model	**LFA** single trajectory
Prior best PG	$\lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert\, t_{\text{mix}}^3$	$\lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert\, t_{\text{mix}}^5$
Ours	$(\lvert\mathcal{S}\rvert\, t_{\text{mix}}^3 + \lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert\, t_{\text{mix}}^2)\, \bar{\chi}^3$	$d\, t_{\text{mix}}^3\, \bar{\chi}^3$
Our critic lower bound	$\lvert\mathcal{S}\rvert\, t_{\text{mix}}^3 + \lvert\mathcal{S}\rvert\lvert\mathcal{A}\rvert\, t_{\text{mix}}^2$	$d\, t_{\text{mix}}^3$

Prior: in expectation, random iterate. Ours: high probability, last iterate.

Paper, code, slides and blog post:
enesarda.com/papers/average-reward-pmd

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